

Experiment # 06

Implementation of Image Histogram and Equalization

Objective

1. To implement image histogram
2. To implement histogram equalization

Software

- Python Idle 3.7 or 3.8.5
- Spyder

Theory

Histogram equalization:

Histogram of an image shows the frequency of different intensities values present in the image. This gives a clear idea of what intensities dominate the image. **Histogram equalization** is a technique that uses this information to enhance the contrast using the probability of a certain pixel to occur.

Steps:

Suppose we have a 3-bit image ($L = 8$) of size 64x64

- a) Calculate number of pixels per pixel value.

Pixel Value (r_k)	0	1	2	3	4	5	6	7
No of Pixels (n_k)	790	1023	850	656	329	245	122	81

- b) Calculate probability density function $PDF = n_k / \text{size of the image}$

No of Pixels (n_k)	790	1023	850	656	329	245	122	81
PDF	0.19	0.25	0.21	0.16	0.08	0.06	0.03	0.02

- c) Calculate cumulative density function $CDF = \text{sum of } n_k \text{ from } 0 - k$

PDF	0.19	0.25	0.21	0.16	0.08	0.06	0.03	0.02
CDF	0.19	0.44	0.65	0.81	0.89	0.95	0.98	1

d) Calculate transformation function by multiplying CDF with $(L - 1)$ and round of it.

CDF	0.19	0.44	0.65	0.81	0.89	0.95	0.98	1
TF	1.33	3.08	4.55	5.67	6.23	6.65	6.86	7
TF (s_k)	1	3	5	6	6	7	7	7

e) Replace original Pixel values with the s_k values

Pixel Value (r_k)	0	1	2	3	4	5	6	7
TF (s_k)	1	3	5	6	6	7	7	7

The Image obtained will be Histogram Equalized Image with high contrast.

Some Useful Commands:

Importing matplotlib: `import matplotlib.pyplot as plt`

1. To plot a simple plot using matplotlib: **`plt.plot( my_data)`**
2. For histogram calculation:
`numpy.histogram(Img, bins, range=None, normed=None, weights=None, density=None)`
3. For histogram equalization: **`cv2.equalizeHist(Img)`**
4. For label along x axis: **`plt.xlabel ( 'Some cooked up data')`**
5. For label along y axis: **`plt.ylabel ( 'Some value')`**
6. To show the graph: **`plt.show()`**

Lab Tasks:

1. Apply Histogram equalization on the given image by applying above steps.
 - i. Calculate the histogram of the image save it as Figure_1.jpg.
 - ii. Calculate probability density function (PDF) from the histogram and save it as Figure_2.jpg. $PDF = H/(R*C)$. Where H is the Histogram and R and C is the number of Rows and Columns of the image respectively.
 - iii. Calculate cumulative density function (CDF) and save it as Figure_3.jpg.

- iv. Multiply the Cumulative PDF with 255 to find the transformation function then save it as Figure_4.jpg
 - v. From the transformation function, replace the gray levels of the image to create contrast enhanced (histogram equalized) image and save it as Figure_5.jpg
 - vi. Calculate the histogram of the output image save it as Figure_6.jpg.
- 2.** For the given image
- i. Calculate histogram of the given image using build-in functions of numpy or OpenCV.
 - ii. Calculate histogram equalization of the given image using build-in functions of numpy or OpenCV and display your output.